

A Unified Picture of Episodic AGN Feedback

Abstract

Active Galactic Nuclei are observed to release significant amounts of energy through feedback processes, including relativistic jets, which quench the Star Formation Rate in galaxies. This paper aims to distinguish the host galaxy properties that drive enhanced energy release. A combined analysis of the LOFAR LoTSS value-added radio survey and the GSWLC-2 SDSS value-added catalog is conducted to retrieve a sample of 40,476 sources. We investigate the relations between Luminosity (at 144 MHz), Mass, Radius, and Star Formation Rate. Our results show a unified picture for density-driven feedback, wherein cycles of activity dominate the radiative luminosity behavior of AGN galaxies.

1 Introduction

1.1 Background Physics

Supermassive Black Holes (SMBHs) residing at the cores of massive galaxies are frequently observed to channel massive quantities of kinetic and thermal energy into their surrounding galactic and intracluster environments (Blandford et al. (2019)). Galaxies hosting these active, accreting black holes are referred to as Active Galactic Nuclei (AGN). A subset of these systems launch powerful, collimated relativistic jets that can propagate across hundreds of kiloparsecs. That is, jets are produced as narrow beams of relativistic plasma that remains constrained as the material propagates through space. As these jets traverse into the ambient halo of hot gas, they undergo hydrodynamic deceleration through shocks, depositing energy in the form of heat and providing a critical feedback mechanism that regulates gas cooling in the host galaxy.

Despite the very large amount of energy released from accreting SMBH that couples to surrounding gas (feedback energy budget associated with an AGN), it is not known how this energy is allocated between radiation, relativistic jets, sub-relativistic winds, and other phenomena. Figure 1, retrieved from NASA (2024), depicts the aforementioned feedback processes. As illustrated, winds are ionized outflows of gas driven outward from the vicinity of the SMBH. In addition, jets inflate cavities from shocks propagated in the inter-galactic medium, which enhances the measured value of a galaxy's radius.

It is well established that AGN feedback has a regulatory effect on the growth of both the SMBH and the host galaxy. In particular, the energy transferred from AGN outflows into surrounding gas in the form of heat leads to the quenching of Star Formation Rate (SFR), as demonstrated robustly by both modern cosmological simulations and observational studies (see Huško et al. (2024) and ref-

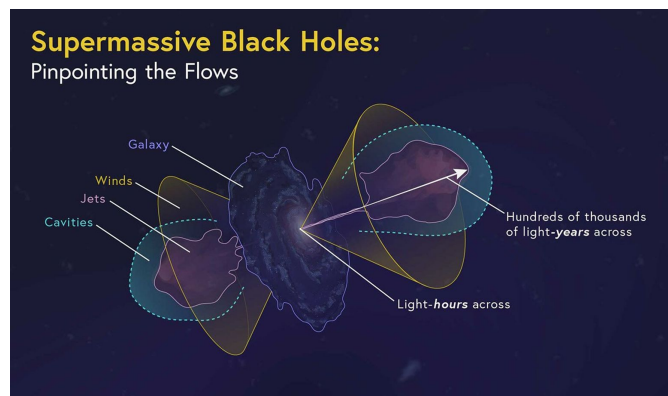


Figure 1: Schematic figure depicting possible feedback processes where the AGN energy budget is allocated.

erences therein). Low-frequency radio observations (e.g., at 144 MHz) serve as an exceptional probe for this feedback, directly tracing the synchrotron emission from relativistic electrons accelerated within these extended jet structures (Harris & Krawczynski (2006)).

1.2 Hypothesis and Objectives

While it is well known that AGN feedback leads to a diminished formation of stars (SFR quenching), the precise physical mechanism governing this suppression remains largely unknown. In this work, two possible explanations are explored:

1. **Density-Driven Accretion:** Host galaxies with higher density produce more AGN power. This can be attributed to more material being present in the same amount of volume, driving more frequent and energetic outbursts.
2. **Enhanced Jet Coupling Efficiency:** Environments with higher density or higher metallicity, both

in the Galaxy and surrounding Cluster of Galaxies, increase jet coupling efficiency. As the jet deposits a higher fraction of energy into the surrounding gas, it leads to an enhanced quenching of SFR.

It is important to mention that no single one of these effects will be the only source of SFR quenching, but rather a complex interplay between them contributes to the resultant observations.

1.3 Emission Mechanisms and Multi-Wavelength Diagnostics

1.3.1 Radio Emission

To empirically test these hypotheses, it is necessary to decouple the observational signatures of the active nucleus from those of the host galaxy’s stellar population. AGN jets are fundamentally composed of a plasma of ultra-relativistic leptons (primarily electrons and positrons) accelerated and collimated at relativistic velocities by strong magnetic fields anchored near the SMBH (Blandford et al. (2019)). Historically, for a variety of extended jet systems, the exact physical mechanisms governing their broadband radiative losses remained an open debate. While non-thermal synchrotron radiation is universally accepted as the driver behind low-frequency emissions, high-energy X-ray emission from large-scale jets was frequently attributed to the Inverse Compton scattering of Cosmic Microwave Background photons (IC/CMB) Harris & Krawczynski (2006). However, crucial gamma-ray observations from the *Fermi* Large Area Telescope have significantly constrained these models. Notably, Meyer et al. (2015) ruled out the IC/CMB framework for the large-scale quasar jets PKS 0637-752 and 3C 273 by demonstrating that the high-energy electron population required to produce X-rays via IC/CMB would inevitably produce a gamma-ray flux that violates observed upper limits. Figure 2¹ illustrates the aforementioned discussion, where IC/CMB emission far exceeds the constraints imposed by Fermi.

By ruling out an independent IC/CMB component, such studies reinforce that the broadband emission of these jets is intimately tied to a continuous, extended population of relativistic leptons. These particles undergo diffusive shock acceleration (first-order Fermi acceleration) along the jet axis, establishing a differential particle energy spectrum that obeys a strict power-law dependence:

$$\frac{dN}{dE} \propto E^{-p} \quad (1)$$

where $N(E)$ represents the electron number density as a function of energy E , and p is the particle injection index. Under the influence of the jet’s ambient magnetic field,

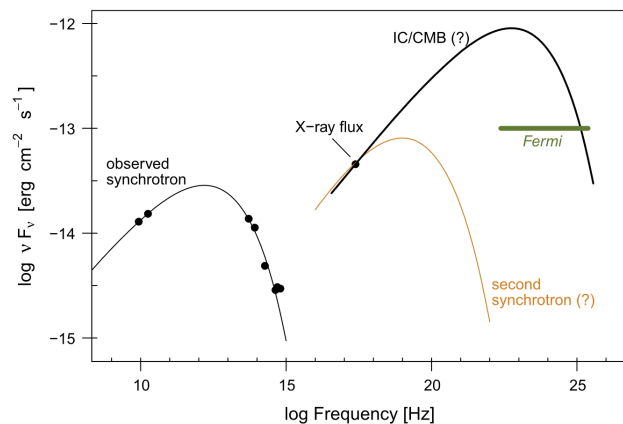


Figure 2: Snippet from Figure 2 in Meyer et al. (2015). Depiction of the Fermi test for the X-ray emission mechanism in large-scale jets, containing the radio, optical, and X-ray photometry for the knots (total) in the jet of PKS 0637-752 as black points.

this power-law distribution of kinetic energies produces a highly characteristic synchrotron radiation spectrum. The resulting observable flux density (S_ν) preserves this underlying power-law behavior across frequency space:

$$S_\nu \propto \nu^\alpha \quad (2)$$

where the radiative spectral index α is mathematically coupled to the electron power-law index via the relationship $\alpha = (1 - p)/2$.

In the framework of this study, this continuous power-law dependence is highly significant. Because synchrotron emission dominates the jet’s energetic profile from radio to higher frequencies without severe contamination from IC/CMB mechanisms, low-frequency observations provide an unclouded view of the jet’s historical energy output. Observing at 144 MHz with LOFAR allows us to sample the lower-energy, long-lived end of this electron power-law distribution. Because these lower-energy leptons are less susceptible to rapid synchrotron cooling than their high-energy counterparts, our measured monochromatic radio luminosity (L_{144}) acts as a reliable, integrated proxy for the total non-thermal kinetic power injected by the jet over time. This foundational baseline allows us to confidently map radio luminosity against internal host galaxy stellar mass densities (ρ) to investigate the environmental drivers of AGN feedback.

It is also important to note that low-frequency radio luminosity remains completely unaffected by the thick columns of dust and gas surrounding AGN cores that frequently obscure the optical and ultraviolet bands. Hence, it serves as a clean proxy for the total kinetic power injected by the jet into the ambient medium. However, by exclusively tracking the lower-energy population, this analysis will be partially incomplete. This effect becomes evident from Figure 2, where another high-energy pop-

¹This snapshot corresponds to Figure 2 in Meyer et al. (2015).

ulation might be dissipating energy in frequencies other than radio.

1.3.2 Optical Surveys and Value-Added Catalogs

In parallel, characterizing the host galaxy requires robust measurements of its overall stellar mass (M_*) and instantaneous Star Formation Rate (SFR). While radio observations isolate the non-thermal emission from relativistic jets, the properties of the host galaxy’s stellar population are primarily encoded in the optical regime, mapped extensively by wide-field spectroscopic campaigns like the Sloan Digital Sky Survey (SDSS). In active star-forming regions, the ultraviolet radiation from short-lived, massive stars ionizes the surrounding interstellar gas, producing prominent optical emission lines such as the Balmer series ($H\alpha$ and $H\beta$). From the ratios of these emission lines, star formation can be determined.

1.4 Observational Approach and Sample Assembly

To systematically investigate these feedback mechanisms across a statistically significant population, wide-area multi-wavelength surveys are essential. This study leverages data from the LOFAR Two-metre Sky Survey (LoTSS), a deep, high-resolution survey conducted with the Low-Frequency Array (LOFAR) operating primarily in the 110–240 MHz band for its High Band Antennas (HBA) ?. Specifically, LoTSS provides unprecedented sensitivity and a spatial resolution of approximately $6''$ at 144 MHz, mapping cosmic star formation and AGN activity without being severely impacted by dust attenuation.

To isolate the radio emission originating from AGN activity from that produced by routine star-forming processes, these low-frequency data must be combined with robust measures of integrated star formation. For this purpose, we cross-match the radio catalog with the Second GALEX-SDSS-WISE Distributed Luminosity Catalog (GSWLC-2), which provides robust stellar masses and dust-corrected star formation rates derived from UV-optical-IR spectral energy distribution (SED) fitting. By utilizing aggregate physical parameters across tens of thousands of galaxies, we map the distribution of radio luminosity against host galaxy stellar density. This wide-field statistical approach allows us to disentangle intrinsic astrophysical correlations from instrumental selection effects.

2 Methodology

A unified framework for the Python analysis detailed below can be found in [Carles Fonseca’s GitHub](#).

2.1 Data Retrieval and Parameter Selection

The data utilized in this study were compiled by cross-matching the LOFAR Two-metre Sky Survey Data Release catalog (LoTSS; J/A+A/678/A151) with the Second GALEX-SDSS-WISE Distributed Luminosity Catalog (GSWLC-2; J/ApJ/859/11) via the VizieR database interface. The cross-matching pipeline was implemented in Python using the `astroquery.vizier` and `astropy.table` libraries.

For simplicity, a value-added catalog of LOFAR LoTSS is analyzed [Hardcastle et al. \(2023\)](#). This allows for a simple cross-referencing of observations in radio and optical frequencies by using their unique Sloan Digital Sky Survey (SDSS) spectroscopic identifiers: the spectroscopic plate number, the fiber ID, and the Modified Julian Date. An inner join is performed based on these parameters. While individually LoTSS retrieved 296,921 sources, and GSWLC-2 retrieved 361,328 sources, the successfully cross-catalog matched sources were 40,476. The analysis detailed below is performed exclusively on these matched sources.

To maximize efficiency and minimize computational memory load, a targeted parsing array is assembled for each source catalog. For the LoTSS database, the requested array is defined as:

```
lotss_columns_parse = ['PlateSDSS', 'FiberSDSS',
                        'MJDSDSS', 'Ftot', 'zphot', 'logMass', 'Type',
                        'r50', 'CompSize']
```

Here, `PlateSDSS`, `FiberSDSS`, and `MJDSDSS` represent the foundational Sloan Digital Sky Survey (SDSS) spectroscopic tracking attributes (Plate number, Fiber ID, and Modified Julian Date), which serve as our absolute spatial coordinates for catalog cross-matching. The remaining attributes dictate core physical parameters:

- **Ftot** tracks the total integrated radio flux density at 144 MHz (used to derive the jet’s radio power L_{144})
- **zphot** supplies the photometric redshift required to determine the cosmological luminosity distance
- **logMass** measures the base-10 logarithm of the integrated stellar mass ($\log_{10}(M_*/M_\odot)$)
- **Type** catalogs the morphological surface profile class used to evaluate sub-population dynamics and structural contamination
- **r50** records the angular half-light radius in arcseconds, which serves as the physical benchmark for calculating internal volume densities (ρ)
- **CompSize** measures the angular composite size of the radio source in arcseconds, which is converted via

the cosmological scale at each source’s redshift into a physical size in kiloparsecs (`kpcSize`) to quantify the spatial extent of the radio jet structure.

On the other hand, the features parsed from the GSWLC-2 catalog were restricted to:

```
WISE_columns_parse = ['Plate', 'Fiber', 'MJD',
                        'logSFR']
```

where `Plate`, `Fiber`, and `MJD` provide the corresponding SDSS identifiers (renamed dynamically upon ingestion to align with the LoTSS schema to execute a clean inner table join), and `logSFR` represents the base-10 logarithm of the integrated, dust-corrected instantaneous Star Formation Rate ($\log_{10}(\text{SFR}/M_{\odot} \text{ yr}^{-1})$).

2.2 Data Filtering and Quality Cuts

To construct a reliable sample for physical analysis, strict quality filters were applied to the matched dataset. Galaxies containing missing entries for key parameters, specifically the log star formation rate ($\log \text{SFR}$) and log stellar mass ($\log M_*$), were discarded. This automated null-filtering pass removed 15,844 incomplete rows, leaving a clean, highly reliable sample of 24,632 galaxies for subsequent calculations.

2.3 Derivation of Physical Parameters

Using the filtered catalog parameters, several intrinsic physical properties were computed assuming a flat Λ CDM cosmological model with a Hubble constant $H_0 = 67.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and matter density parameter $\Omega_{m,0} = 0.308$, implemented via `astropy.cosmology.FlatLambdaCDM`.

2.3.1 144 MHz Radio Luminosity (L_{144})

The total integrated flux density at 144 MHz (`Ftot`), provided in units of mJy, was first converted to SI flux density units ($1 \text{ mJy} = 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1}$). The monochromatic radio luminosity (L_{144} , in units of W Hz^{-1}) was evaluated using the standard cosmological relation:

$$L_{144} = 4\pi d_L^2(z) \cdot F_{\text{tot}} \cdot K(z) \quad (3)$$

where $d_L(z)$ is the cosmological luminosity distance in meters computed from the photometric redshift (`zphot`). To account for cosmic expansion shifting the rest-frame frequency, a standard radio K -correction was applied:

$$K(z) = (1+z)^{\alpha-1} \quad (4)$$

where $\alpha = -0.7$ represents the characteristic synchrotron spectral index ($S_{\nu} \propto \nu^{\alpha}$) typical of extended radio galaxies. The final values were cast into dex format ($\log_{10} L_{144}$).

2.3.2 Physical Size and Mean Stellar Mass Density (ρ)

The spatial extent of each galaxy was parameterized by its angular half-light radius (`r50`), measured in arcseconds. The angular scale was converted to a proper physical radius (R_{50} , in units of cm) using the cosmological proper scale at the galaxy’s redshift:

$$R_{50} = r_{50} \times [\text{kpc proper per arcsec}(z)] \times (3.086 \times 10^{21} \text{ cm/kpc}) \quad (5)$$

Assuming a uniform spherical distribution as a first-order approximation, the physical volume V enclosing half of the galaxy’s light was defined as:

$$V = \frac{4}{3} \pi R_{50}^3 \quad (6)$$

The total stellar mass M_* was retrieved by converting the catalog parameter `logMass` out of logarithmic form ($M_* = 10^{\log M_*}$, in units of $M_{\odot}$). Finally, the mean internal stellar mass density (ρ) within the half-light radius was calculated:

$$\rho = \frac{M_*}{V} \quad [\text{M}_{\odot} \text{ cm}^{-3}] \quad (7)$$

A safety mask was integrated into the pipeline to enforce $V > 0$, avoiding numerical divisions by zero for compact or unresolved sources.

2.4 Classification of Active Galactic Nuclei via Radio Excess

In galaxies without an active nucleus, low-frequency radio emission is dominated by free-free and synchrotron radiation originating from star-forming regions (specifically, supernova remnants accelerating cosmic rays). Because this star-formation-driven radio luminosity scales linearly with the total SFR, AGN can be identified by their prominent “radio excess” above this empirical baseline [Hardcastle et al. \(2019\)](#).

An empirical cutoff line was established in the $\log L_{144}$ versus $\log \text{SFR}$ parameter space to separate the two populations. The boundary was optimized to track the top edge of the star-forming main sequence, defined by the linear relation:

$$\log_{10} L_{144, \text{cutoff}} = 2.0 \times \log_{10} \text{SFR} + 23.5 \quad (8)$$

Sources positioned strictly above this threshold ($\log_{10} L_{144} > \log_{10} L_{144, \text{cutoff}}$) were flagged as radio-excess AGN (`cutoff = True`), while sources below were categorized as routine star-forming galaxies (SFGs).

To quantitatively evaluate the reliability of this empirical cut, a morphological contamination rate was defined. Late-type disk galaxies and regular resolved sources classified in the catalog under the structural designations

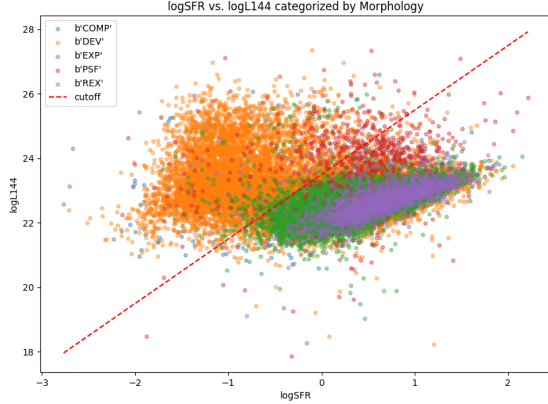


Figure 3: Classification of sample galaxies via radio excess. The red dashed diagonal line marks the optimized empirical selection boundary defined by the linear relation $\log_{10} L_{144, \text{cutoff}} = 2.0 \times \log_{10} \text{SFR} + 23.5$. Galaxies above this cutoff are classified as AGN, and those below as SFG.

EXP (exponential disk profile) or REX (residual/regular expanded) are robustly assumed to be predominantly star-forming systems rather than powerful radio-loud AGN. The contamination rate was computed as the fraction of these morphological types that “spilled” above the selection boundary relative to the total number of radio-excess sources:

$$\text{Contamination Rate} = \frac{N_{\text{EXP, above}} + N_{\text{REX, above}}}{N_{\text{Total, above}}} \times 100\% \quad (9)$$

This calculation yielded an remarkably low contamination rate of 3.59%, confirming that the selection boundary effectively suppresses false-positive AGN classifications while isolating a clean sample of active nuclei. Figure 3 illustrates the cutoff line, as well as the color-coded types of galaxies found in this classification. While the method for obtaining the cutoff selection is biased, the contamination rate and visual examination reassure the validity of criteria.

3 Results and Discussion

3.1 Scaling Relations and Multi-Panel Analysis

To evaluate the environmental, structural, and evolutionary drivers behind active galactic nuclei (AGN) feedback, we examine the distribution of the 144 MHz radio luminosity (L_{144}) against three foundational macroscopic properties of our sample: the integrated host stellar mass ($\log M_*$), the stellar mass density (ρ), and the physical composite size of the radio emission structure (kpcSize).

By evaluating these multi-wavelength parameters exclusively for the radio-excess AGN sub-population isolated via our empirical star-forming boundary cutoff, we map out how the central engine interacts with its large-scale environment over its lifespan.

Figure 4 presents these scaling relations as a three-panel composite figure, illustrating the physical and structural footprints of the active systems.

3.2 Mass-Driven Jet Energetics

From a physical standpoint, more massive host galaxies possess substantially deeper gravitational potential wells, providing a larger overall reservoir of matter. In addition, empirical black hole scaling relationships ($M_{\text{SMBH}} - M_{\text{bulge}}$) mandate that these massive galaxies² harbor more massive central SMBHs (Kormendy & Ho (2013)). The massive central engine drives higher accretion onto the center of the galaxy. In turn, this fuels the injection of plasma required to power magnetic flux and launch powerful relativistic jets, elevating the synchrotron output measured at 144 MHz. This positive correlation between the host galaxy’s stellar mass $\log M_*$ and the radio power $\log L_{144}$ is observed in Figure 4a.

According to the Density-Driven Accretion in Section 1.2 we expect galaxies with higher density to produce more AGN power. It follows from the above paragraph that massive galaxies correlate to higher power output, so the Density-Driven Hypothesis indicates that higher mass should be positively correlated with higher density. In Figure 5 we plot Luminosity vs. host mass color coded by density. The results contradict the Hypothesis, indicating that galaxies with higher density have lower mass and radio power output. While this result is confusing at the moment, a possible explanation for this behavior is detailed in Section 3.5.

3.3 The Density-Size Column Structure and Jet Lifetimes

In Figure 4c we observe a clear scaling relationship between the physical composite size of the radio structure *kpcSize* and its Luminosity in the 144 MHz frequency. This distribution shows significantly less scatter and “vertical pillar” behavior compared to Figures 4b and 4a. This linear scaling relationship serves as a structural timeline for the episodic life-cycle of AGN activity. As an AGN becomes active, the injection of powerful jets drives hydrodynamical expansion and inflates radio lobes. Consequently, the physical size of the radio galaxy acts as a reliable proxy for the elapsed time of the current feedback episode rather than any intrinsic property of the

²Here, mass refers to the stellar mass of an elliptical galaxies. Nuances apply to other cases.

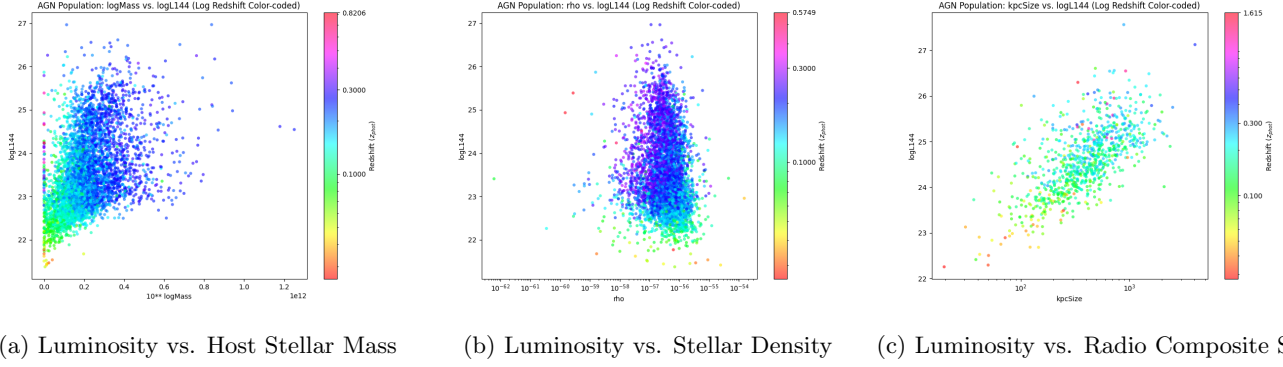


Figure 4: Multi-panel distribution of the 144 MHz radio luminosity ($\log_{10} L_{144}$) against host galaxy and jet parameters for the radio-excess AGN sample, color-coded by photometric redshift (z_{phot}). Panel (a) shows the positive correlation between radio power and integrated stellar mass ($\log M_*$). Panel (b) reveals a distinct, vertical pillar-like distribution mapping out the core environment density. Panel (c) highlights a broad linear scaling between the physical composite radio size (kpcSize) and radio luminosity, tracking the age and expansion history of the jet structures.

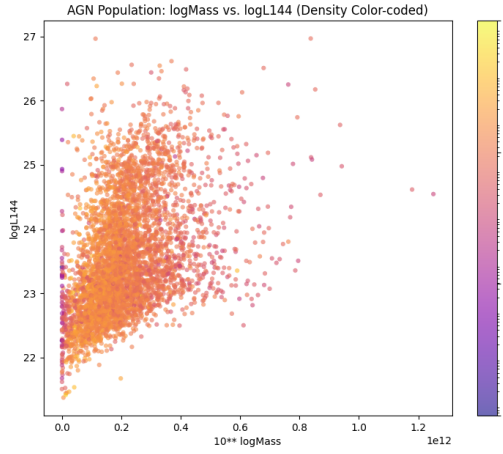


Figure 5: 144 MHz radio luminosity ($\log_{10} L_{144}$) against host galaxy Mass for the radio-excess AGN sample, color-coded by density

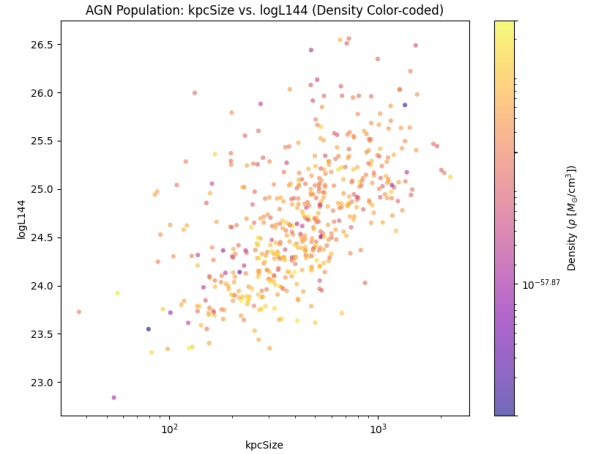


Figure 6: 144 MHz radio luminosity ($\log_{10} L_{144}$) against host galaxy size for the radio-excess AGN sample, color-coded by density

host galaxy itself. Figure 6, which maps radio luminosity vs. galaxy size color coded by density, shows no underlying trend in mass, confirming that hydrodynamic expansion, rather than host galaxy mass scaling, is the primary driver of the radio structure's spatial growth.

Conversely, when mapping radio luminosity against the host internal stellar mass density ρ in Figure 4b, vertical column- or pillar-shaped distributions spanning orders of magnitude in radio power are observed for small steps in density. An inverse correlation between density ρ and luminosity $\log L_{144}$ is observed. This relationship appears, at first glance, to contradict the Density-Driven Hypothesis, wherein higher densities produce more AGN power. Nevertheless, Figure 4b can be understood as a snapshot of a galaxy at a given moment in its lifespan.

Hence, galaxies with higher luminosities, corresponding to higher AGN feedback output, would have undergone hydrodynamic expansion which decreases their density. In turn, higher AGN power is proportional to lower densities as expected from the Density-Driven Hypothesis. While this result is consistent, it is by no means a direct proof of its validity.

In summary, over time, AGN feedback drives a major hydrodynamic expansion of the surrounding gas and interstellar medium, effectively diminishing the density of the galaxy. Consequently, as a jet is created and propagated, its physical footprint in the form of kpcSize expands outward, while the bulk internal density drops sharply. The column-like distribution is therefore a snapshot of active systems observed at varying stages of this

feedback-driven expansion loop.

3.4 Redshift Trends and Cosmic Selection Effects

An intriguing feature of the composite size scaling in Figure 4c is the redshift distribution within the sample. The largest physical radio structures are predominantly found at higher photometric redshifts (corresponding to a younger age of the universe), while the most compact radio sources reside heavily at lower redshifts (closer to the modern epoch).

This behavior is primarily a reflection of cosmological selection effects and survey volume limits. Because this sample samples the nearby universe up to a maximum redshift of $z \approx 1.615$, the volume enclosed at higher redshifts is vastly larger than the local volume. Since powerful, giant radio galaxies are statistically rare, they are preferentially detected within the massive volume sampled at higher redshifts. Furthermore, galaxies in the higher-redshift universe possess significantly higher cold gas fractions, providing a more robust ambient medium to confine and sustain extended synchrotron emission structures before they undergo severe radiative cooling, allowing them to grow to larger visible scales.

3.5 A Unified Picture of Episodic AGN Feedback

Synthesizing these observations yields a self-consistent, unified model of episodic AGN feedback driven by a self-regulating evolutionary loop:

1. **The Accretion Phase:** A high-mass host galaxy provides a large mass reservoir within a compact, high-density environment. This material funnels into the central engine, igniting a high-luminosity radio jet event.
2. **The Feedback and Expansion Phase:** The relativistic jet propagates outward over kiloparsec scales, transferring kinetic energy to the gas. This drives a major outward expansion, simultaneously increasing the physical radio footprint ($kpcSize$) and lowering the bulk density (ρ) of the core environment.
3. **The Quenching and Starvation Phase:** This ongoing expansion heats and evacuates the interstellar medium, quenching the host's Star Formation Rate (SFR). Crucially, this process also deprives the central SMBH of its immediate fuel supply, slowing accretion and causing the active jet phase to subside.
4. **The Recurrence Cycle:** Over extended cosmological timescales, the displaced gas cools and collapses back into the gravitational potential well of the host

galaxy. This gradual re-accumulation slowly rebuilds the central density reservoir until a critical threshold is crossed, re-igniting the central engine and initiating a new cycle of episodic AGN feedback.

Evidence for the cyclical nature of jet feedback and the recurrence cycle is provided below. Nevertheless, it is important to mention that this is a well-established phenomenon (Huško et al. (2024)) and is therefore included in this Unified Picture.

3.6 Binned 2D Distributions and Evolutionary Trajectories

To isolate the structural conditions required to trigger active galactic nuclei (AGN) and track their subsequent physical evolution, the sample is stratified into four equal-distribution stellar mass slices, spanning from low-mass dwarfs to massive elliptical galaxies. By examining the 2D histograms ($\log_{10} \rho$ vs. $\log_{10} L_{144}$) in Figure 7 and the corresponding joint scatter-marginal plots in Figure 8, we can contrast the stable, steady-state evolution of SFGs against the highly dynamic, cyclical life-cycles of active systems.

3.6.1 The Star-Forming Baseline: Structural Equilibrium

Across all four stellar mass slices in Figure 7, the SFG population (represented in the right-hand columns) exhibits a remarkably stable, symmetric distribution forming tight, concentric, elliptical contours centered around a distinct population locus. The rounded, oval shape indicates that a galaxy's physical structure and its stellar population (originating radiation) are deeply connected.

This structural uniformity implies that non-active galaxies govern their baryonic content through smooth, long-term equilibrium paths. As a galaxy grows in stellar mass from Slice 1 (7a) through Slice 4 (7d), its mean internal stellar density (ρ) follows a clean, predictable 1D Gaussian distribution as evidenced in the scatterplot across all mass bins in Figure 8. The low-frequency radio emission in these systems scales directly as a linear, passive tracker of star-driven particle acceleration, maintaining a narrow vertical spread at any given density slice.

The reason thousands of galaxies are found in the center of these ellipses is attributed to a self-correcting balance (Kennicutt (1998)). Whenever hydrogen gas falls into a region of the galaxy, it becomes denser and sparks a rapid wave of star formation. Yet, this sudden formation rapidly leads to powerful supernova explosions, thereby ejecting excess gas and quenching the SFR. Hence, any given galaxy that drifts too far from the average baseline is naturally regulated to converge in the middle of the

ellipse. Ultimately, Figure 7 represents a statistical snapshot of galaxies living in stable environments, fluctuating around a point of physical equilibrium. In addition, the vertical height is governed by the "Star-Forming Main Sequence" (Noeske et al. (2007)), an empirical rule showing that galaxies have a natural limit for SFR based on their mass.

3.6.2 AGN Triggering Thresholds and the Arrow-Like Trajectory

In stark contrast to the SFGs, the radio-excess AGN population displays a highly asymmetric distribution that varies drastically as a function of host galaxy mass. In Mass Slice 1 of Figure 7 ($\log_{10}(M_*) \in [3.80, 10.43]$), powerful radio AGN are almost entirely absent, confirming that low-mass potential wells rarely host the massive central supermassive black holes (SMBHs) or stable fueling reservoirs required to ignite relativistic jets. However, as the host mass increases through Slices 2, 3, and 4, an active population rapidly emerges, shifting decisively toward **higher central stellar densities** ($\log_{10} \rho \gtrsim -57 \text{ M}_\odot \text{ cm}^{-3}$) **and luminosities** ($\log L_{144} \gtrsim 22 \text{ W Hz}^{-1}$). This pattern demonstrates that a highly concentrated host core environment is a mandatory physical prerequisite for launching high-power radio outbursts.

For the massive hosts in Slices 3 and 4 (Figures 7c and 7d), the 2D count map reveals a striking, elongated "arrow-like" or *wedge-shaped distribution* pointing from the upper-left down toward a high-density convergence zone at the lower-right. This morphological feature provides a direct statistical visualization of a feedback-driven life-cycle caught in transition:

- **The Arrow Head (Triggering and Peak Fueling):** The dense base or "head" of the arrow, located at the maximum density and moderate-to-high radio power ($\log_{10}(L_{144}) \approx 22.8$), represents the onset of the active event. Here, substantial gas fractions have collapsed into the core, driving up the initial accretion rate and lingering long enough to create a high concentration of sources in parameter space. Since AGN events are the most extreme phenomena in the cosmos, it should be expected that their effects vary vastly between galaxies. Therefore, the bright collection of sources at the "head" of the arrow can only be attribute to a quiescent AGN period.
- **Outward Excursions and Jumps:** The extended shaft of the arrow, stretching toward the upper left, represents active systems undergoing rapid hydrodynamic expansion. Large vertical "jumps" along the y -axis ($\log_{10} L_{144}$) depict distinct, episodic epochs of nuclear activity where the central engine undergoes highly energetic outbursts of synchrotron injection.
- **The Cooling and Starvation Tail:** As the

launched jet continues to push outward, clearing the central core and inflating the radio composite size (kpcSize), the local density drops sharply. This pushes the system into the far-left tail of the arrow, characterized by exceptionally low densities ($\log_{10} \rho \lesssim -57.8$) and declining radio luminosities. This domain captures the starved, cooling phase of the cycle where local star formation is quenched and the jet eventually fades due to fuel exhaustion.

These trends are uncovered primarily due to the tight stellar mass binning applied to our sample. Because each sub-plot isolates a narrow interval of mass, the observed variation across orders of magnitude in density cannot be attributed to structural scaling laws between different galaxy masses. Instead, it is directly attributable to the localized hydrodynamic expansion of individual host galaxies. The left-hand panels of Figure 7 explicitly demonstrate that, relative to the "Arrow Head" triggering locus, an increase in radio luminosity strongly correlates with decrease in bulk density, providing direct observational support for the feedback-driven expansion loop outlined in previous sections. Nevertheless, it is unclear whether the "cooling and starvation trail" represents a true evolutionary feature observed in this plot or whether it arises from human expectation bias. While the connection between these low-density, low-luminosity outliers and the death of a single jet episode remains a highly plausible theoretical trajectory, it requires further modeling to be robustly confirmed.

3.6.3 Disentangling Cosmic Evolution from Localized Feedback

A critical challenge in interpreting these broad vertical columns is determining whether the variation in radio luminosity for a given density is driven by the physical aging of individual jets or is merely an artifact of observing different cosmic epochs. In other words, the "pillars" observed in Figure 4b might be attributable to an observational bias. That is, faint radio galaxies cannot be observed at higher redshifts, since their dim light would not record significant counts in observational instruments. In addition, bright objects might be statistically rare, yet detected in abundance at higher redshifts because the interferometers are instruments are probing a wider volume in space. This issue is resolved by analyzing the multi-panel joint scatter plots color-coded by photometric redshift (z_{phot}) in Figure 8.

Looking at the distributions attached to the vertical axis (L_{144}), a clear, large-scale redshift gradient is observed. The lower-luminosity boundary is dominated by local, low-redshift sources (darker hues), while the highly luminous upper boundary is heavily populated by high-redshift sources (lighter hues). This vertical sorting is a combined consequence of Malmquist bias—where

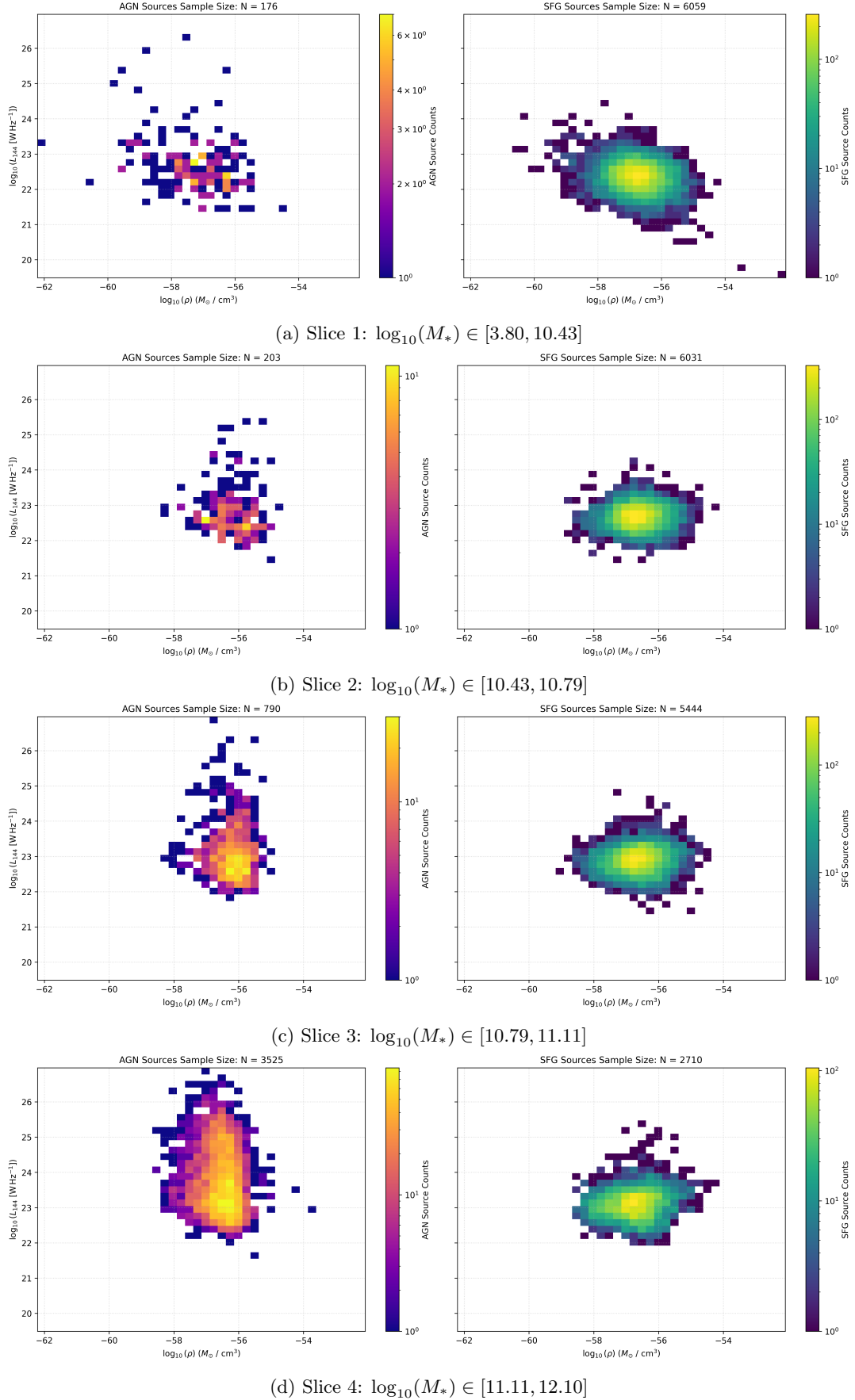


Figure 7: 2D histograms of $\log_{10}(\rho)$ vs $\log_{10}(L_{144})$ for AGN and SFG sources, split across four equal-distribution stellar mass slices.

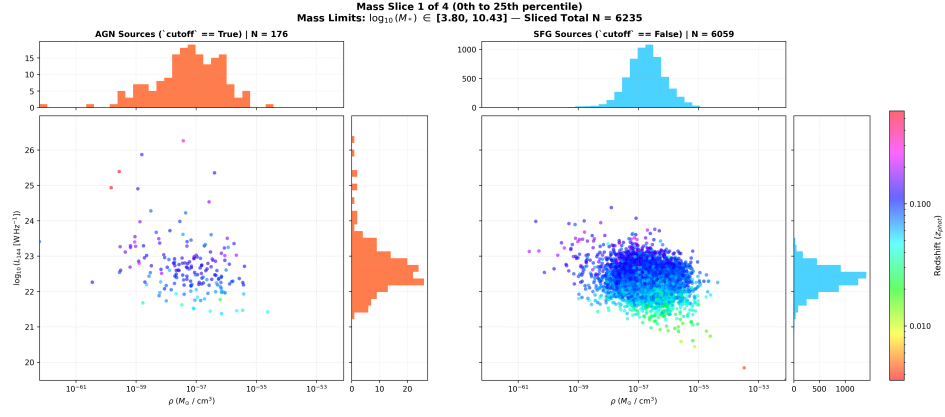
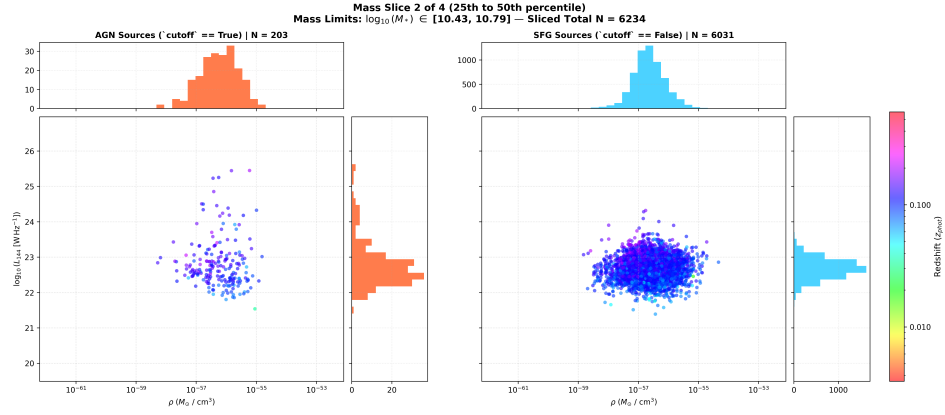
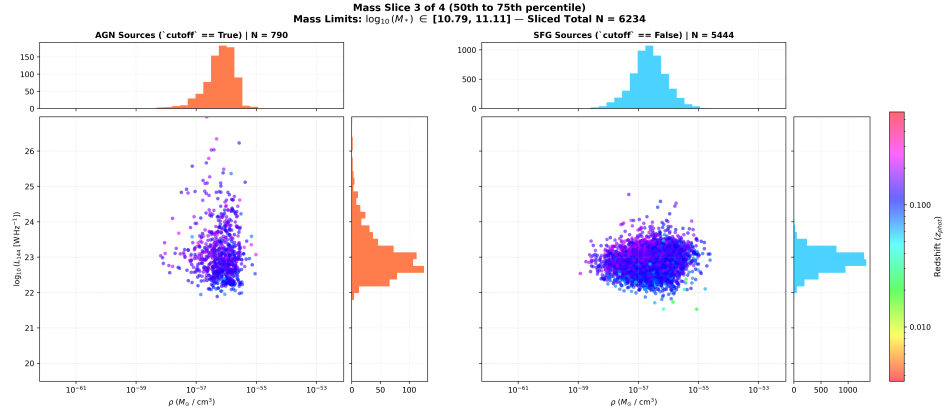
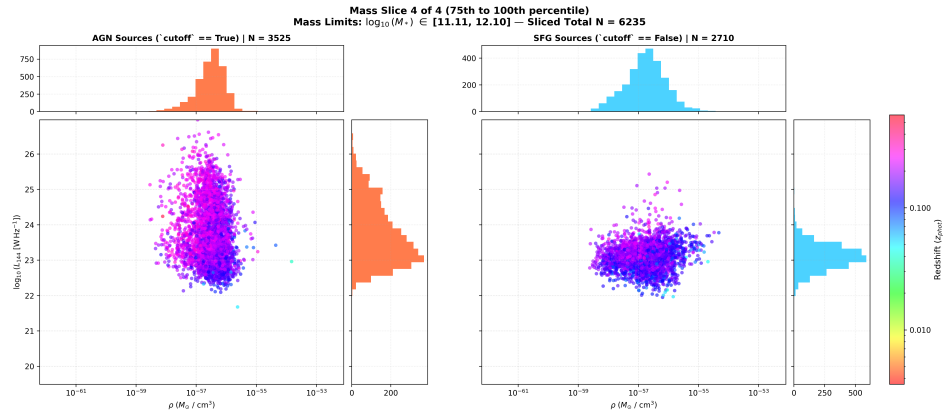
(a) Slice 1: $\log_{10}(M_*) \in [3.80, 10.43]$ (b) Slice 2: $\log_{10}(M_*) \in [10.43, 10.79]$ (c) Slice 3: $\log_{10}(M_*) \in [10.79, 11.11]$ (d) Slice 4: $\log_{10}(M_*) \in [11.11, 12.10]$

Figure 8: Multi-panel joint scatter plots (ρ vs $\log_{10}(L_{144})$) color-coded by photometric redshift with corresponding side-panel marginal histograms for both AGN and SFG populations across four distinct stellar mass quantiles.

distant, faint radio sources fall below the LOFAR flux threshold—and the vast survey volumes scanned at higher redshifts, which statistically favor catching rare, ultra-powerful radio outbursts.

Crucially, however, the horizontal axis marginal distributions (ρ) and the internal scatter distributions within the high-mass slices (8c and 8d) reveal a thorough **mixing and overlapping of redshifts at intermediate densities**. For a given redshift slice, galaxies do not cluster at a single static coordinate; instead, they span the entire density-size pillar structure. This physical blending confirms that the broad distributions and “arrow-like” trajectories provide a robust statistical snapshot of localized, self-regulating episodic feedback loops operating continuously across all observed epochs of the universe.

4 Conclusion

4.1 Summary of Background and Methodology

AGN feedback driven by SMBH is a well established driver for quenching SFR in massive host galaxies, albeit the environmental mechanisms regulating this energy production and deposition remain actively debated [Huško et al. \(2024\)](#). In this work, we hypothesize that highly concentrated density environments provide more fuel to spark energetic outburst, and also that dense or high-metallicity mediums increase the jet coupling efficiency onto surrounding gas.

A statistical analysis matching 44 MHz low-frequency radio observations from the LOFAR Two-metre Sky Survey (LoTSS) with dust-corrected physical parameters from the Second GALEX-SDSS-WISE Distributed Luminosity Catalog (GSWLC-2) via the Vizier interface is conducted to evaluate these theories ?.

4.2 Novel Structural Insights and the Unified Model

While radio power (L_{144}) exhibits a expected positive correlation with integrated host stellar mass (M_*) due to basic scaling laws, mapping L_{144} against the mean internal stellar mass density (ρ) reveals a novel structural feature: a distinct vertical pillar-like or column distribution where radio luminosity spans multiple orders of magnitude across narrow steps in density. By examining these columns alongside the physical composite size of the radio structures (`kpcSize`), the results demonstrate that this distribution is fundamentally governed by the lifecycle and aging of individual jet outbursts.

These trends converge in a unified model of episodic AGN feedback operating via a dynamic, self-regulating

loop:

1. **Accretion Phase:** A massive host galaxy clusters gas into a compact, high-density core, fueling the central engine and triggering a high-luminosity radio jet outburst.
2. **Feedback and Expansion Phase:** The relativistic jet propagates outward over kiloparsec scales, transferring immense kinetic energy to the gas. This drives hydrodynamic expansion that inflates the visible radio footprint (`kpcSize`) while causing the host’s core internal density (ρ) to drop sharply.
3. **Quenching and Starvation Phase:** The ongoing expansion heats and evacuates the gas reservoir, simultaneously quenching the local Star Formation Rate (SFR) and starving the central engine of fuel, causing the active jet phase to fade.
4. **Recurrence Cycle:** Over extended timescales, the displaced gas cools and collapses back into the host’s gravitational potential well, slowly rebuilding the high-density reservoir until a critical threshold is crossed, re-igniting the central engine to initiate a new active epoch.

This cyclical lifecycle is visually captured in massive host galaxies by a striking, elongated “arrow-like” trajectory in the left column of Figure 7, where the dense head maps peak nuclear triggering, vertical excursions track episodic synchrotron injections, and the extended left-hand tail isolates the starved, low-density cooling phase.

Nevertheless, it is important to note that the complete “Quenching and Starvation” and “Recurrence Cycle” phases are not directly captured by our analysis alone. Instead, this interpretation is inferred based on the theoretical frameworks and numerical simulations of preceding studies (e.g. [Huško et al. \(2024\)](#)). Nonetheless, this idealized model aligns closely with our observational findings, which serves as a robust indicator of consistency.

4.3 Honest Appraisal of Hypotheses

An honest evaluation indicates that further work remains to prove our hypotheses:

- **Hypothesis 1 (Density-Driven Accretion):** In this work, we find that a highly concentrated host environment ($\log_{10} \rho \geq -57 \text{ M}_\odot \text{cm}^{-3}$) operates as a mandatory physical prerequisite to ignite radio outbursts. Nevertheless, the correlation between the most luminous sources and their corresponding densities has not been formally proven. While the “arrow-like” shape of the 2D Histogram in Figure 7 suggests the build up of density before outbursts can occur, it is unclear whether this population ultimately powers the brightest events.

- **Hypothesis 2 (Enhanced Jet Coupling Efficiency):** This hypothesis cannot be fully or directly answered using the macroscopic parameters tracked in this study. While our unified model confirms that kinetic energy is successfully transferred to expand and heat the host medium, the aggregate stellar mass density (ρ) and integrated low-frequency radio luminosity (L_{144}) are insufficient proxies to isolate small-scale jet-gas interaction physics or track localized metallicity variations.

4.4 Strengths, Limitations, and Cosmic Implications

The primary strength of this research lies in its vast statistical scale, which disentangles astrophysical correlations from individual instrumental biases, combined with dust-insensitive 144 MHz observations that provide an unclouded proxy for the jet’s integrated kinetic power. Conversely, limitations include spectral incompleteness, as high-energy particle sub-populations may dissipate energy outside the radio band, and geometric simplifications stemming from assuming a uniform spherical mass distribution over complex galactic morphologies.

In the broader context of galaxy evolution, this research reinforces a critical paradigm shift: massive systems do not evolve along smooth, steady-state equilibrium paths. Instead, their structural evolution is fundamentally non-linear, dynamic, and episodic. Black hole feedback acts as the primary self-regulating cosmic mechanism that breaks structural equilibrium, dictating how and when a galaxy transitions between active star-bursting phases and mature, quiescent states.

4.5 Future Steps and Societal Importance

To comprehensively answer our second hypothesis regarding localized jet coupling efficiency and composition, a critical future step will be to complement this large-scale statistical study with spatially resolved Spectral Energy Distribution (SED) fitting across the extent of individual host galaxies. Following this study with resolved SED mapping will enable to directly probe the internal structural composition of the galaxy and see the multi-phase material interacting inside and along the path of the jet, explicitly uncovering how localized ambient environments influence the efficiency of energy deposition.

While studying distant active galaxies can appear abstract, this investigation carries substantial importance for non-experts and society as a whole. On a fundamental level, understanding AGN feedback provides insights into the most energetic events in the cosmos. Furthermore, the methods developed to observe, calibrate, and analyze

are directly transferrable into daily human technological advances. In addition, understanding the interaction of high-energy particles elucidates the nature of matter and its composition. Hence, by resolving these cosmic structures humanity might better understand the arrangement and origin of matter, leading to endless technological and scientific developments.

5 Assignment Wrapper

Research Practice 98T Assignment Submission Wrapper

Assignment: Complete Final Draft

Name: Carles Fonseca Mauri

Date: June 4, 2026

Academic honesty statement: This First Draft of the Data Project Assignment has relied heavily on previous Data Factsheets for the Research Practice class, which were in part motivated by Claire Williams’s guidance. In addition, Artificial Intelligence has been used to understand the data and other research papers, albeit the majority of the writing and analysis has been carried out independently. Finally, I received valuable help from my instructor, Claire Williams, who assisted me in analyzing the results for this assignment. Furthermore, I would like to acknowledge and express my gratitude for the helpful comments and insightful feedback received during the review process from both Claire Williams and an anonymous reviewer, which significantly improved the clarity and quality of this draft.

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